
Algorithm 1: vulnLabel(vuln_ranges, commit_graph): label the global commit graph with vulnerability ranges.

Input : vuln_ranges: list of vulnerability ranges; commit_graph: global commit graph

Output: commits_vulns: Dictionary mapping commit IDs to sets of vulnerability ranges

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(1) commit_to_vulnerability_range ← empty dictionary
(2) foreach range in vuln_ranges do
(3)   patched_commit ← empty dictionary ; // Currents commit status, True = Patched, False = Vulnerable
(4)   stack ← empty list
(5)   foreach introduction_commit_ids in range do // Step 1 : Add all introduction commits to the stack
(6)     stack.add(introduction_commit_ids)
(7)     patched_commit.put(introduction_commit_ids, False)
(8)   end
(9)   while stack is not empty do ; // Step 2 : Process stack element and their child
(10)
(11)     current_commit ← stack.pop()
(12)     foreach child_commit of current_commit in commit_graph do
(13)       is_child_patched ← (child_commit is not an introduction event) and
                           (child_commit is a fix commit or
                           child_commit is a limit commit or
                           child_commit already marked as patched or
                           current_commit marked as patched or
                           current_commit is a last_affected commit)
(14)       // Update patched_commit if child_commit not already processed by other parent or the status of child_commit has changed
(15)       if patched_commit [child_commit] != is_child_patched then
(16)         patched_commit.put(child_commit, is_child_patched)
(17)         stack.put(child_commit)
(18)       end
(19)     end
(20)   end
(21)   if range contains a limit commit = then // Step 3 : Handle Limit commit
(22)     // Filter non-patched visited commits that are not in the linear path between the limit and introduced commits
(23)     patched_rev_filtered ← empty dictionary
(24)     stack ← empty list
(25)     foreach limit_commit in range do
(26)       stack.add(limit_commit)
(27)     end
(28)     while stack is not empty do
(29)       current_commit ← stack.pop()
(30)       foreach parent_commit of current_commit in commit_graph do
(31)         if parent_commit in patched_commit and not patched then
(32)           patched_rev_filtered.put(parent_commit, False)
(33)         end
(34)         stack.add(parent_commit)
(35)       end
(36)     end
(37)     patched_commit ← patched_rev_filtered
(38)   end
(39)   foreach commit in patched_commit where not patched do // Step 4 : Update commit_to_vulnerability_range
(40)     commit_to_vulnerability_range [commit].add(range)
(41)   end
(42) end
(43) return commit_to_vulnerability_range
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